Where Does Long-Term Agent Memory Actually Fail? A Lifecycle-Boundary Attribution Protocol with Oracle Controls

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Abstract

Long-term-memory agents are evaluated end-to-end: a wrong answer is scored wrong, and nothing is learned about where the pipeline broke. We argue that this measurement gap hides the dominant failure mode of memory systems, and we fix the instrument. We present Lifecycle-Boundary Attribution, a frozen protocol that attributes every wrong answer to a specific boundary of a nine-stage memory lifecycle (conversation → formation → storage → evolution → retrieval → reranking → context → reasoning → answer). Three design choices make the attribution reliable: an anti-circular evidence-sufficiency gate that compares gold evidence against the benchmark reference answer rather than the model's own prediction; a semantic decision layer in which lexical metrics never enter attribution; and control conditions—Real Memory, Oracle Memory (gold injected as formed memories into an independent engine), and Oracle Context (gold injected directly into the context)—that separate retrieval-side from reasoning-side failures. Running the protocol on two long-term-memory benchmarks (LongMemEval-s and LoCoMo) with paired statistics, two answer models, two formation extractors, and a judge-reliability study with author adjudication, we find that the dominant boundary is formation, not retrieval: real memory loses a large fraction of gold evidence before it ever reaches the store, and once evidence reaches the final context the answer model rarely fails. The boundary shifts downstream on long conversations and moves again when the extractor is swapped, which is why an attribution protocol rather than a single bottleneck claim is the transferable contribution. The protocol, the measurement-validity fixes it required, and the full per-question traces are released as an open artifact.

1 Introduction

Agent memory systems—retrieval-augmented stores, consolidation pipelines, reflective memories—are increasingly the substrate of long-horizon LLM agents, and they are evaluated almost exclusively end-to-end: answer a question over a long conversation, score the answer. That score conflates every way the system can fail. A missing fact, a superseded belief, a retrieval miss, a truncated context window, and a reasoning slip all produce the same observable: a wrong answer.

Prior work on memory failures either taxonomizes errors post-hoc from traces or reports aggregate accuracy, and the attribution instruments underneath carry three measurement problems that we document and fix here: (i) lexical scorers (exact match, token F1) are used as ground truth for correctness, conflating representation failures with reasoning failures; (ii) evidence matches by lineage (a fact citing a source turn) are treated as content preservation; (iii) "oracle" controls are underspecified—injecting gold evidence into the context and injecting it into memory are different interventions, and equating them misattributes retrieval failures to reasoning.

Contributions.

1. A frozen attribution protocol with an anti-circular evidence gate. Lifecycle-Boundary Attribution labels every wrong answer with the earliest lifecycle boundary at which gold evidence is not fully present. Before any failure label is assigned, an E0_evidence_insufficient gate asks whether the benchmark's own reference answer is recoverable from the gold evidence—comparing against the reference rather than the model's prediction,

which breaks the "wrong answer → missing evidence → blame memory" circularity. All thresholds are frozen before any large-scale run (§3).

2. Measurement-validity fixes, each with regression tests. Lexical scores never enter attribution (they conflate representation failures with reasoning failures); provenance matches require content preservation; semantic evidence matching requires faithful representation rather than topical overlap; deterministic judge channels are narrowed to quote/date rules; embedding-backend failures fail loudly instead of silently degrading. Each fix ships with regression tests, including two calibration-period false negatives that pinned the bugs.

3. A judge-reliability study with author adjudication. We measure inter-judge agreement (Cohen's κ) between two judge models on a seeded subsample, adjudicate every disagreement against gold evidence under published guidelines, validate the adjudication against a third judge family re-judging the same answers strictly blind, and report how the headline numbers move under the adjudicated verdicts (Study 2E). To our knowledge this is the first adjudicated judge-reliability measurement in long-term memory evaluation.

4. An empirical study on two benchmarks, two answer models, and two formation extractors, with paired statistics (exact McNemar, paired bootstrap CIs), per-question-type breakdowns, and a cross-model replication. The dominant failure boundary is formation; reasoning failures (F6 candidates) are rare and mostly control failures of the oracle conditions. An extractor ablation (Study 2F) shows the boundary is a property of the formation stage, not of one rule-based implementation: with an LLM-driven extractor, formation recall rises from 0.236 to 0.838 and the oracle-memory advantage collapses from +17.3 points to zero, relocating the residual bottleneck to benchmark evidence sufficiency and reasoning.

The three-condition design (Real Memory / Oracle Memory / Oracle Context) is the measurement instrument these contributions run on, not a contribution in itself: oracle-style controls have become common in memory evaluation (Long et al., 2026; Sidik and Rokach, 2026). What is new is the protocol around them—the anti-circular gate, the frozen thresholds, the regression-tested validity fixes, and the adjudicated judge study.

2 Related Work

Long-term memory benchmarks. LongMemEval (Wu et al., 2025) evaluates chat assistants over long histories with question types that stress extraction, temporal reasoning, and knowledge updates; LoCoMo (Maharana et al., 2024) evaluates very long multi-session conversations. We use both as evidence-boundary substrates and treat their gold evidence spans as the lifecycle signal.

Memory architectures. RAG-style stores (Lewis et al., 2020; Robertson and Zaragoza, 2009), consolidation and reflection pipelines (Park et al., 2023; Packer et al., 2023; Shinn et al., 2023), and recent agent-memory systems (Chhikara et al., 2025; Xu et al., 2025; Liu et al., 2024b) optimize recall or answer accuracy. These works report what the system remembers or answers; ours reports where along the lifecycle the evidence was lost, on a concrete open implementation, so the protocol's claims are checkable rather than asserted.

Failure analysis and attribution. Error taxonomies for agents and RAG failure analyses (Zhang et al., 2024) typically require human trace reading, and long-context degradation studies (Liu et al., 2024a) characterize attention-side loss rather than memory-pipeline loss. Our protocol automates the lifecycle attribution while keeping the human in the loop for judge reliability (§5).

Diagnosing where memory fails (2024–2026). A first wave of automated failure-attribution work targets multi-agent systems: Who&When localizes the responsible agent and the decisive failure step (Zhang et al., 2025b); AgenTracer uses counterfactual replay and fault injection to label failure traces

(Zhang et al., 2025a); AgentDebug classifies errors into memory, reflection, planning, action, and system-level modules (Zhu et al., 2025). Two 2026 preprints probe long-term memory systems specifically, and their conclusions differ from ours in an instructive way. MemTrace measures per knowledge point across 13 system configurations and four paradigms, and reports that the dominant bottleneck is evidence use, not retrieval: when systems fail, the evidence was retrievable ten times more often than it was missing (Long et al., 2026). MEMTIER analyzes retrieval bottlenecks in a tiered memory architecture on LongMemEval-S with Qwen2.5-7B, reporting that structured pre-population lifts single-session recall to 0.686–0.714 (Sidik and Rokach, 2026). We read the three-way divergence—MemTrace's "evidence use", MEMTIER's "retrieval", and our formation dominance—not as a contradiction but as the central motivation for a protocol: the dominant boundary moves with the system under test and the workload shape, so no single pooled score can locate it. Our Study 2F isolates exactly this variable by swapping the formation extractor inside one fixed pipeline—the oracle-memory advantage collapses from +17.3 points to zero when the extractor is replaced, while Study 2C shows the boundary shifting downstream on longer conversations (§5). Oracle-style controls are now common in this literature (Long et al., 2026; Sidik and Rokach, 2026); the contribution we claim is not the oracle construction itself but the attribution protocol around it.

Evaluation methodology. LLM-as-judge and semantic evaluation (Zheng et al., 2023; Liu et al., 2023); judge reliability and self-preference bias (Kadavath et al., 2022); uncertainty and abstention (Kadavath et al., 2022). We measure inter-judge agreement explicitly (Cohen's κ, Study 2E) and keep the judge model separate from the answer model in the cross-model design (Study 2D). For condition comparisons we use exact McNemar tests (McNemar, 1947) with paired bootstrap CIs (Efron, 1979), and treat questions within one conversation as correlated (Dietterich, 1998), reporting per-type breakdowns rather than pooling silently.

3 Protocol: Lifecycle-Boundary Attribution

Lifecycle. Conversation → Formation → Storage → Evolution → Retrieval → Reranking → Context → Reasoning → Answer. Each arrow is a measurable boundary with its own artifact set.

Labels. F1_formation (evidence never formed), F2_storage (formed but not stored), F3_evolution (stored but superseded/dropped by governance), F4_retrieval (in store but not retrieved), F5_reranking (retrieved but dropped by reranking), F6_reasoning _candidate (all evidence reached context, answer still wrong), E0_evidence_insufficient (gold evidence insufficient for the reference answer), eval_semantic_uncertain (judge undecided), rc_reasoning_control_failure (oracle control answered wrong).

Decision order. (1) Semantic verdict on the model answer; correct → no failure. (2) Undecided → eval_semantic_uncertain. (3) E0 gate: is the gold evidence sufficient for the benchmark reference answer (never the model prediction)? Insufficient → E0_evidence_insufficient. (4) Earliest lifecycle stage whose artifact set does not fully contain the gold evidence → F1–F5. (5) All stages reached and answer still wrong → F6_reasoning _candidate.

Evidence matching tiers. Exact text → provenance with content-preservation check (shared content vocabulary $\geq$ 50% of the smaller side and $\geq \min(2, |gold|)$ shared tokens; rejected matches are reported as provenance_only) → semantic (faithful-representation containment at 0.8, either direction, same absolute floor). Raw-token overlap is never a match: two same-topic different-event evidence items ("…the park on 9 May 2023" vs "…the park on 7 May 2023") score 0.56 under the legacy rule and are conflated, which is one mechanism behind inflated legacy recall.

Unmeasured stages. Stages a condition does not measure carry measured: false and are skipped, never read as perfect recall. Oracle Context, for example, bypasses the pipeline; its unmeasured stages are excluded from stage

recall.

Embedding-backend discipline. Embeddings serve retrieval only; attribution runs on content. A failing embedding request is retried and then fails the record loudly (hash fallback is opt-in offline mode); inputs exceeding the backend context window are deterministically halved; every run is verified dimension-clean afterwards (§4).

4 Experimental Design

System under test. A concrete open implementation: a formation extractor, an embedding store with dense retrieval and reranking, context assembly, and a local answer model (qwen2.5:7b, temperature 0; gemma3:12b cross-check), bge-m3 embeddings. The formation extractor has two arms behind one interface: a deterministic rule-based analyzer (facts, relations, events, rules, profiles) and an LLM-driven extractor (qwen2.5:7b, Study 2F). Everything downstream of formation— governance, storage, retrieval, reranking, context assembly—is identical across arms, so the extractor is the only varying component. The rule arm makes no LLM calls at all; only answer generation, semantic judging, embeddings, and the 2F extraction arm hit the model server.

Conditions. Real Memory: full pipeline over the conversation. Oracle Memory: gold evidence injected as formed active facts into an independent engine, then the full retrieval/reranking/context pipeline. Oracle Context: gold evidence placed directly into the answer context, bypassing the pipeline. The three conditions partition failure responsibility: Real vs. Oracle Memory isolates the memory-side loss; Oracle Memory vs. Oracle Context isolates context assembly; Oracle Context's own wrong answers are rc_reasoning_control_failure.

Datasets and sampling. LongMemEval-s (n=10 full haystack, protocol replication); LongMemEval oracle subset (n=50, two-stage stratified by conversation and question type, largest-remainder allocation, per-type minimum of one, fixed seed); LoCoMo (n=50, stratified by conversation). Question-level subsets are committed with sampling recipes; conversations are joined at load time from the local full data; raw benchmark data is not redistributed.

Metrics and statistics. Per-condition semantic accuracy (judge verdict; lexical EM/F1 reported in parallel, never as attribution input); per-stage recall; failure-label distribution. Condition comparisons use exact McNemar tests on discordant pairs and paired bootstrap percentile CIs (10,000 resamples). Judge reliability uses Cohen's κ between the primary judge and a second judge model re-judging the same stored answers.

Hypotheses. H1: Real Memory loses a significant fraction of gold evidence before context; F1/F4/F5 dominate over F6 candidates. H2: Oracle Memory $\gg$ Oracle Context accuracy implies retrieval/context assembly, not reasoning, is the bottleneck. H3: Real-vs-oracle inversions (oracle wrong, real right) are control failures, not evidence of reasoning strength.

5 Results

Study 2A (protocol replication, n=10, full haystack). Three-condition independence holds in the strict-mode rerun: Real Memory 7/10 semantic accuracy, Oracle Memory 10/10, Oracle Context 10/10. The real-vs-oracle gap is directionally large (+0.300) but not significant at n=10 (exact McNemar b=3, c=0, p=0.25; bootstrap CI $[+0.000, +0.600]$)—this study exists to validate the protocol, not to estimate effects. First losses: F1_formation×8, F3_evolution×1, context×1; stage recall formation 0.200, storage 1.000, evolution 0.100, retrieval 1.000, rerank 1.000, context 0.650. The E0 gate finds all 10 items evidence-sufficient. Calibration-period false negatives (e47becba, 58bf7951) are explained by the formation boundary.

Study 2B (primary, n=50). Table 1 reports the strict-mode rerun (post-C8 fix; embedding backend verified clean: 119 databases, 7,000 vectors, all 1024-dim). Semantic accuracy (uncertain verdicts excluded from the denominator) is 66.0% for Real Memory (31/47), 83.3% for Oracle Memory (40/48), and 75.0% for Oracle Context (36/48). Real Memory loses gold evidence at the formation boundary in 45/50 cases (90%): formation recall 0.236

(95% CI [0.147, 0.331]) and evolution recall 0.236, against storage 1.0, retrieval 1.0, rerank 0.987 (CI [0.967, 1.000]), and context 0.695 (CI [0.598, 0.791]; bootstrap over questions, 10,000 resamples). The formation/evolution tie is mechanical: facts that fail formation are also absent from every later stage, so evolution recall inherits the formation deficit rather than adding an independent one. Oracle Memory restores every pre-context stage to 1.0 and reaches 0.979 context recall; its remaining three failures are context-assembly, not formation. The E0 gate finds 40/50 benchmark items carry sufficient evidence, 8 insufficient (reported as E0_evidence_insufficient, not as memory failures), and 2 uncertain (excluded). Among attributable answer failures, F1_formation dominates (9), followed by E0_evidence_insufficient (7) and eval_semantic_uncertain (3); zero F6_reasoning candidates occur, because gold evidence never survives formation.

Real versus Oracle Memory is significant: exact McNemar on discordant pairs b=10, c=2, p=0.0386; paired bootstrap (10,000 resamples) 95% CI on the accuracy difference [+0.043, +0.319]. Real versus Oracle Context is marginal (b=6, c=1, p=0.1250, CI [+0.000, +0.217]), consistent with H2: repairing evidence availability (Oracle Memory) closes most of the gap, while bypassing the pipeline (Oracle Context) closes less. Per question type (Table 2), the formation dominance is broadest on multi-session and temporal-reasoning items (4 F1_formation each) and weakest on single-session-user/assistant items (1.0 accuracy in all conditions). The pre-fix run (real 58.3%, oracle memory 72.9%, oracle context 70.2%) is retained in the artifact as an audit reference only.

Study 2C (cross-benchmark, LoCoMo n=50). LoCoMo questions are embedded in much longer multi-session conversations (∼600 messages per question across ∼28 sessions, versus ∼115K-token histories in Study 2B), so this study tests whether the formation-dominant boundary survives a change of benchmark and workload shape. It does, with one informative shift. Semantic accuracy is 26/50 = 52.0% (real) vs. 36/50 = 72.0% (oracle memory) vs. 37/50 = 74.0% (oracle context); both oracle contrasts are significant (real vs. oracle memory: exact McNemar b=15, c=5, p=0.0414, paired bootstrap 95% CI [+0.040, +0.360]; real vs. oracle context: b=16, c=5, p=0.0266, CI [+0.040, +0.380]). First loss is again concentrated at formation (F1_formation 46/50; none 1, F4_retrieval 2, F5_reranking 1), and the answer-failure distribution is F1_formation×18, E0_evidence_insufficient×3, F4_retrieval×3 (E0 gate: 47 sufficient, 3 insufficient). The shift is on the retrieval side: with the same rule extractor, retrieval recall drops from 1.000 (2B) to 0.608 (CI [0.483, 0.728]) and reranking from 0.987 to 0.527 (CI [0.398, 0.655]) once conversations grow to hundreds of messages, so F4_retrieval losses appear for the first time (3/24 attributable). Formation recall is even lower than in 2B (0.097, CI [0.028, 0.182]): long conversations stress the extractor further. The boundary moves with the workload, but it has not left the pre-context stages.

The shift concerns the retrieval side. In Study 2B the real-memory stage recall was 1.0 at retrieval and 0.99 at reranking; on LoCoMo it drops to 0.61 and 0.53, and retrieval-side failures appear for the first time (F4_retrieval 12.5% of answer failures). Long, topically diffuse conversations thus move part of the loss downstream of formation: the extractor still loses the majority of evidence (formation recall 0.097), but the subset it does store is no longer always retrievable. Lexical false negatives are even stronger here (lexical 28.0% vs. semantic 52.0% on real answers), reinforcing

	Real	Oc. Mem.	Oc. Ctx.
Semantic accuracy*	66.0%	83.3%	75.0%
Lexical accuracy†	38.0%	44.0%	50.0%
Formation recall	0.236	1.000	n/a
Storage recall	1.000	1.000	n/a
Evolution recall	0.236	1.000	n/a
Retrieval recall	1.000	1.000	n/a
Rerank recall	0.987	1.000	n/a
Context recall	0.695	0.979	1.000
First loss = F1	45/50	0/50	n/a

Table 1: Study 2B (LongMemEval oracle subset, n=50). F1 = F1_formation. *Semantic accuracy = 31/47, 40/48, 36/48 (uncertain verdicts excluded from the denominator). †Lexical accuracy = 19/50, 22/50, 25/50, reported for reference only and never an attribution input.

Question type	n	Real	Oracle Mem.	Oracle Ctx.	Attributed failures (real)
single-session-user	7	1.000	1.000	1.000	EVAL×1
single-session-assistant	6	1.000	1.000	1.000	—
single-session-preference	3	0.667	0.667	0.667	E0×1
knowledge-update	8	0.750	0.875	0.875	F1×1, E0×1
multi-session	13	0.455	0.636	0.545	F1×4, E0×2, EVAL×2
temporal-reasoning	13	0.462	0.846	0.615	F1×4, E0×3

Table 2: Study 2B semantic accuracy by question type. Failure labels abbreviated: F1 = F1_formation, E0 = E0_evidence_insufficient, EVAL = eval_semantic_uncertain. Formation failures concentrate in multi-session and temporal-reasoning items.

category	n	Real	Oracle Mem.	Oracle Ctx.	failures (real)
multi-hop	10	0.500	0.700	0.800	F1×5
temporal	10	0.300	0.600	0.600	E0×3, F1×2, F4×2
open-domain	9	0.333	0.444	0.444	F1×6
single-hop	21	0.714	0.905	0.905	F1×5, F4×1

Table 3: Study 2C semantic accuracy by LoCoMo category (1=multi-hop, 2=temporal, 3=open-domain, 4=single-hop); label abbreviations as in Table 2, F4 = F4_retrieval. Temporal items are weakest; single-hop items, though best, still lose 5/21 to formation.

that lexical scoring would misattribute most of these failures.

Study 2D (cross-model, gemma3:12b). Replacing the answer model with gemma3:12b (judge unchanged: qwen2.5:7b) tests which parts of the attribution are properties of the memory pipeline rather than of the answer model. The lifecycle side is unchanged exactly as the protocol predicts: real-memory stage recall is identical to Study 2B (formation 0.236, first loss F1_formation 45/50), because extraction never invokes the answer model. The answer side compresses: semantic accuracy is $28/47 = 59.6\%$ (real) vs. $32/49 = 65.3\%$ (oracle memory) vs. $34/49 = 69.4\%$ (oracle context). The real-vs-oracle-memory contrast loses significance (b=7, c=4, p=0.549, CI $[-0.064, +0.213]$)—a weaker answer model leaves less headroom for the oracle advantage to express—while real vs. oracle context remains directional with c=0 (real never beats oracle context; b=5, p=0.063, CI $[+0.021, +0.191]$). One inversion appears in the multi-session type (real 0.500 vs. oracle memory 0.417), consistent with oracle injection occasionally crowding the context. The cross-model design also bounds judge dependence: the judge model is held fixed across 2B and 2D, so the lifecycle labels—formation dominance in particular—cannot be artifacts of judge–answer model co-adaptation, and Study 2E directly measures the judge axis.

Study 2E (judge reliability). Gemma3:12b rejudges the stored answers of a seeded 20-record subsample of Study 2B (seed 20261018; 55 paired verdicts where both judges returned one, 9 more settled by the model-independent deterministic channel). Raw agreement is $43/55 = 78.2\%$, but Cohen's κ over the three-way verdict (correct/incorrect/uncertain) is only 0.158—a prevalence effect whose structure matters for reading the accuracy numbers: the judges are reliable on the consensus majority (41/55 agree-correct) and unreliable only inside a 14-verdict borderline band, concentrated on answers needing a small inference step from the evidence. Because a low κ is actionable only if we know which judge is right, the authors adjudicated all 16 disagreement cases against gold evidence and benchmark reference answers under written guidelines (Appendix D); two preference-type items are unadjudicable and excluded. On the 14 decidable cases the qwen2.5:7b judge matches the adjudicated truth 8/14 times, gemma3:12b 3/14; gemma's errors are 9 over-acceptances and 0 over-rejections (systematic leniency), qwen's are balanced (2/2).

Third-family blind validation of the adjudication. The adjudication is itself a measurement, so we validate it against a third family that cannot have co-adapted with either pri-

mary judge: DeepSeek (deepseek-v4.1-flash) re-judges the same 60 answers under the identical prompt and deterministic rules, strictly blind—it sees only the question, the answer, and the gold evidence; never the primary verdicts, the adjudication, or the fact that rows were disagreement-selected. It reproduces the adjudication on 13/16 cases (13/14 decidable) while agreeing with neither primary judge (9/16 qwen, 4/16 gemma) and in 5 cases sides with the adjudicated truth against the primary judge, so the adjudication is not a restatement of the qwen verdicts. The blind judge is itself miscalibrated ($\kappa = 0.409$ vs. qwen, 0.198 vs. gemma), so it serves only as an independence check, never an accuracy source. Applying the adjudicated verdicts as a sensitivity analysis moves the headline numbers at most 4.2 points (real memory 66.0%→62.5%, Oracle Memory 83.3%→81.6%, Oracle Context 75.0%→79.2%), leaves the condition ordering unchanged, and strengthens the primary contrast (real vs. Oracle Memory: b=11, c=2, p=0.0225). The lifecycle attribution is untouched: stage recall is computed from pipeline artifacts, not verdicts, so judge choice enters only at the failure/no-failure and E0/EVAL boundaries, exactly where we report its measured disagreement.

Study 2F (extractor ablation, n=50). The decisive external-validity control: the same 50 LongMemEval-s oracle questions, the same answer model, judge, embeddings, and lifecycle stages—only the formation extractor changes, from the rule analyzer to an LLM-driven one. The LLM arm issued 354 extraction calls (23 batches unparseable and counted-fallback to the rule extractor, covering 37 messages; zero silent drops) and the formation stage observes 2,262 memory items against 335 for the rule arm on the same questions (6.8×, median per-question ratio). Formation recall rises from 0.236 (CI $[0.147, 0.331]$) to 0.838 (CI $[0.747, 0.918]$), evolution recall from 0.236 to 0.802, and context recall from 0.695 to 0.847; first loss is F1_formation in 13/50 instead of 45/50, and 29/50 questions (was 5/50) now lose no gold evidence at any stage.

The oracle conditions collapse accordingly. Oracle Memory still repairs every pre-context stage to 1.000 (context 0.979, three context-assembly failures as in 2B), but its answer-level advantage over Real Memory disappears: semantic accuracy is 70.8% (34/48) real vs. 71.4% (35/49) oracle memory vs. 70.8% (34/48) oracle context, with real-vs-oracle-memory b=4, c=4, p=1.0000, CI $[-0.125, +0.125]$—against $+0.170$, p=0.0386 under the rule extractor. The attributable failure mix shifts from the evidence side to the evaluation side: F1_formation drops from 9 to 3, E0_evidence_insufficient from 7 to 6, and F6_reasoning (reasoning candidates: sufficient evidence, full context coverage, semantically wrong answer) appears for the first time, 4 cases, alongside 2 eval_semantic_uncertain and 1 F5_reranking. Formation dominance is thus a property of the extractor, not of the pipeline: replacing the extractor moves the boundary out of formation and leaves a residual budget dominated by benchmark evidence sufficiency and reasoning.

6 Analysis

Where memory fails. The stage-recall chain localizes the loss: formation recall is the lowest measured boundary, and context recall is high once evidence is stored. This is an architectural statement: coverage of the extractor, not retrieval quality, is the binding constraint for the systems under test. In Study 2B the pattern is sharp—45/50 real-memory runs lose gold evidence at formation (recall 0.236), while storage, retrieval, and reranking are at or near ceiling (1.000, 1.000, 0.987). Once Oracle Memory injects gold as formed facts, every pre-context stage returns to 1.000 and only three context-assembly failures remain. The deficit is therefore almost entirely upstream of retrieval: improving rankers or context windows cannot recover evidence the extractor never formed.

The boundary moves with workload shape. Study 2C replicates formation dominance on a second benchmark (first loss F1_formation 46/50, recall 0.097) and adds a boundary condition: on long, topically diffuse multi-session conversations (∼600 messages/question), part of the loss migrates downstream of formation. Retrieval-stage recall falls from 1.000 (Study 2B) to 0.608, reranking from 0.987 to 0.527, and F4_retrieval appears among answer fail-

ures (3/24). The attribution protocol makes this shift visible instead of averaging it away: formation remains the primary boundary on both benchmarks, but its share is a property of the workload, not a constant—shorter curated histories lose evidence almost entirely at formation, while long conversational histories additionally stress retrieval and reranking.

The oracle gap is a property of the extractor. The most consequential measurement finding of this paper is that the oracle-memory advantage is not a constant of the pipeline family. Under the rule extractor, Oracle Memory beats Real Memory by +0.170 (p=0.0386); under the LLM extractor, on the same 50 questions, the advantage is +0.000 (p=1.0000, CI $[-0.125, +0.125]$). The mechanism is visible in the stage chain: the rule arm forms 335 memory items and misses 76.4% of gold evidence before storage; the LLM arm forms 2,262 items and misses 16.2%. Once formation recall reaches 0.838, injecting gold as formed facts adds little the extractor has not already captured, so the oracle condition stops discriminating. Two implications follow. First, an oracle ablation run on a single extractor measures the extractor, not the architecture: effect sizes from oracle-style controls are only comparable across systems whose formation stages have been characterized. Second, our headline formation dominance is explicitly bounded—it holds for the rule arm on both benchmarks and is replaced by an evidence-sufficiency/reasoning-dominated residual under the LLM arm, which is the honest scope of the claim and the reason the extractor axis belongs in any replication of this protocol.

Divergent bottleneck claims in the literature. Recent work on long-term memory diagnosis reports different dominant bottlenecks: MemTrace, probing 13 system configurations across four paradigms, concludes that the bottleneck is evidence use rather than retrieval (when systems fail, the evidence was retrievable ten times more often than it was missing) (Long et al., 2026); MEMTIER reports retrieval bottlenecks in a tiered architecture and shows structured pre-population lifting single-session recall to 0.686–0.714 (Sidik and Rokach, 2026). Our formation dominance on a rule-extractor pipeline appears to disagree with both. We read the divergence as the protocol's point rather than its refutation: each study measures a different system under test, and the dominant boundary is a property of that system's formation stage. Study 2F demonstrates this directly inside one pipeline: a rule-based extractor with no recall-oriented training loses most evidence before storage (formation recall 0.236, oracle-memory advantage +17.3 points), an LLM-driven extractor recovers most of it (0.838, advantage +0.0, CI $[-0.125, +0.125]$) and relocates the residual bottleneck to evidence sufficiency and reasoning, and MemTrace's systems retrieve evidence but use it poorly. A pooled accuracy score cannot distinguish these regimes—which is exactly why the attribution instrument, not any single boundary claim, is the transferable contribution.

Why end-to-end accuracy misleads. Lexical scoring produces false negatives on concise correct answers, which the semantic layer resolves; conversely, lineage matches without content preservation inflate recall. Both effects move attributed failures toward the wrong boundary—and both are regression-tested. In Study 2B, lexical accuracy under real memory (38.0%) sits 28 points below the semantic verdict (66.0%): a system evaluated lexically would look retrieval-broken when it is extraction-broken.

What the E0 gate changes. Before any failure label, the gate asks whether the benchmark's own reference answer is recoverable from the gold evidence. In Study 2B, 8/50 items fail this check (and 2 are judge-uncertain), i.e., 20% of the nominal error budget is benchmark evidence insufficiency or judge uncertainty, not memory failure. Protocols that skip this gate silently convert benchmark noise into system failures; we report E0_evidence_insufficient and eval_semantic_uncertain as first-class outcomes.

Oracle inversions. Real-vs-oracle inversions are labeled rc_reasoning_control_failure and reported separately; they bound how much of an oracle deficit is reasoning-side. In Study 2B, two records are correct under Real Memory but wrong under Oracle Memory (c=2):

the oracle pipeline is not uniformly better, and the paired statistic accounts for it. Oracle Context—which bypasses the pipeline—answers 75.0% despite having perfect context, so reasoning on perfect evidence is itself imperfect; this is why H2 is stated as Oracle Memory $\gg$ Oracle Context rather than oracle perfection. The inversion between the two oracle conditions (Oracle Context 75.0% below Oracle Memory 83.3%, with context recall 1.000 in both) is a presentation effect, not a coverage effect: Oracle Memory delivers gold evidence through the same reranking and top-k context assembly the answer model is tuned for, while Oracle Context dumps every gold span into the context at once, diluting attention over the relevant span—the long-context degradation pattern documented for LLMs (Liu et al., 2024a). The practical reading is that "give the model the right evidence" is not a sufficient intervention; how evidence is assembled into the context window is part of the failure surface.

7 Limitations

External validity of the system under test. All conclusions come from one open memory implementation whose formation stage we ablate between a rule-based and an LLM-driven extractor (Study 2F), both deterministic in their storage, governance, retrieval, and reranking stages. Formation-boundary numbers characterize this pipeline family, not all memory systems; LLM-native systems with learned extraction and consolidation (e.g., (Chhikara et al., 2025; Sidik and Rokach, 2026)) may locate their dominant boundary elsewhere, and MemTrace's evidence-use finding on 13 system configurations (Long et al., 2026) is a case in point. The ablation bounds this concern rather than eliminating it: Study 2F shows the boundary relocates (formation 0.236→0.838, oracle advantage +17.3 points→+0.0) when the extractor changes, so the transferable claim is the protocol and the extractor-dependence of the boundary, not any single boundary location. We further report the workload-dependence of the boundary (2B vs. 2C) and two answer models (2B vs. 2D), but a broader system sweep is future work.

Sample size and statistical power. n=50 per benchmark; estimates carry the corresponding uncertainty (reported as paired bootstrap CIs for both accuracy differences and stage recall). Study 2A (n=10) is a protocol-replication check, not an effect-size estimate, and is underpowered by design. Questions within one conversation are correlated (Dietterich, 1998); we report per-type breakdowns and treat pooled p-values as optimistic.

Judge sensitivity. The judge is an LLM. Study 2E measures cross-family judge disagreement and finds a prevalence-inflated raw agreement (κ=0.158) with a systematic leniency asymmetry (gemma3:12b over-accepts). Author adjudication of all 16 disagreements shows the primary judge matching adjudicated truth 8/14 times and quantifies the effect on headline numbers (±4 points; ordering unchanged, primary contrast strengthened). The adjudication is itself validated against a third judge family re-judging the same answers strictly blind (agreement 13/16 with the adjudicated truth, 9/16 with the primary judge), which bounds, but does not eliminate, the residual concern that the adjudicators were not independent humans. E0/EVAL outcome counts remain judge-dependent, and the adjudication covers only the 20-record subsample; full human annotation of all studies is future work.

Benchmark coverage. LoCoMo category semantics were mapped from question inspection and secondary documentation rather than an authoritative legend (the mapping table ships with the artifact). Benchmarks are English conversational QA; temporal and multi-hop question types are represented but not exhaustively. The public artifact must be anonymized for review per the current call.

8 Conclusion

Where does long-term agent memory actually fail? In the system families we measured, the answer moves with the formation stage. Under a rule-based extractor the boundary sits at formation (45/50 runs lose gold evidence before storage); replace the extractor with an LLM-driven one and the same pipeline's boundary relocates downstream, the oracle-memory advantage collapsing from +17.3 points to zero. That mobility, not any single boundary location, answers the title question, and it is

why the transferable contribution is the protocol rather than a boundary claim: Lifecycle-Boundary Attribution turns a wrong answer into a lifecycle attribution with three oracle controls, an anti-circular evidence-sufficiency gate, regression-tested measurement fixes, an adjudicated and blind-validated judge study, and paired statistics—an instrument that stays informative when the boundary moves, as end-to-end accuracy and lineage-only or lexical-only instruments do not.

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A Protocol Pseudocode

The attribution decision order is frozen in docs/PROTOCOL.md and implemented once in evaluation/attribution.py. Lexical scores are computed for reporting only and never enter this procedure.

```
def attribute(answer, gold, ref, stages):
    v = verdict(answer, gold)
    if v == CORRECT:                    # (1)
        return NO_FAILURE
    if v == UNCERTAIN:                  # (2)
        return EVAL_SEMANTIC_UNCERTAIN
    if not sufficient(gold, ref):       # (3) E0 gate
        return E0_EVIDENCE_INSUFFICIENT
    for stage in [FORMATION,            # (4) F1-F5
                  STORAGE, EVOLUTION,
                  RETRIEVAL, RERANKING]:
        if not covers(stages[stage], gold):
            return FAILURE_LABEL[stage]
    return F6_REASONING_CANDIDATE       # (5)
```

Frozen thresholds: content-preservation check — shared content vocabulary $\geq 50\%$ of the smaller side and $\geq \min(2, |gold|)$ shared tokens; semantic tier — directional containment ≥ 0.8 with the same absolute floor; deterministic judge support R1 — shared span covers ≥ 0.8 of the candidate (verbatim-quote cases only); deterministic judge support R2 — pure date/relative expressions that resolve to the same date as timestamped evidence. provenance_only matches are reported as diagnostics, never as stage coverage.

B Per-Question Attribution Tables

Every study writes one JSON object per question to comparison.jsonl with: the question and benchmark answer; the gold evidence spans (with ids and timestamps); per-condition answers; the semantic verdicts and judge metadata; the per-stage artifact coverage used by the attribution decision; the assigned failure label; and the code commit of the run. The per-question tables of all studies are released as an artifact alongside the paper; Table 1 and Table 2 aggregate them.

C Judge Prompts and Normalization

The semantic judge is prompted to decide whether the model answer is correct given the gold evidence, and to answer yes/no/uncertain with a one-line rationale; the judge never sees the benchmark reference answer (the E0 gate owns that comparison) and never sees lexical scores. Two deterministic channels repair judge output before it is accepted, both deliberately narrow: R1 lexical accepts a no verdict only when the candidate is essentially a verbatim quote of the evidence (shared span covers ≥ 0.8 of the candidate); R2 temporal accepts pure date/relative expressions that resolve to the same date as timestamped evidence. Anything wider passed wrong answers during calibration ("went to the park" style candidates), which is why the thresholds are frozen and regression-tested. The cross-model judge (Study 2E) uses the same prompt with gemma3:12b; agreement is reported as Cohen's κ on the shared subsample.

D Adjudication Guidelines

The 16 disagreement cases of Study 2E were adjudicated by the authors against the gold evidence and the benchmark reference answer, under the following written criteria, applied identically to every case and blind to which judge produced which verdict:

1. Decidable only on evidence. A case is decidable only if the gold evidence plus the benchmark reference answer determine the verdict. If the reference answer is too vague to grade (two preference-type items), the case is marked unadjudicable and excluded from the strict counts rather than forced.

2. Correct iff fully supported. An answer is correct only if every load-bearing claim it makes is supported by the cited gold evidence and consistent with the reference answer. Concise, paraphrased, and answer-equivalent forms count as correct.

3. No unlicensed inference. An answer that requires a step the evidence does not license (a different entity/event assumed identical, a superlative supported by a single item, a small arithmetic step the reference answer does not itself make) is incorrect. Borderline cases are labeled borderline, not silently forced.

4. Temporal resolution. Relative expressions (yesterday, last Saturday) resolve against evidence timestamps before grading.

5. Independence check. The adjudicated verdicts were then validated against a third judge family (DeepSeek, deepseek-v4.1-flash) re-judging the same answers strictly blind; agreement is reported in §5 and per-case rationales are in the artifact (results/study2e_judge_gemma/adjudication_report.txt).

E Stratification Recipes

Study subsets are built by experiments/build_study_set.py. Primary stratum: conversation (or session group) so that no single conversation dominates and within-conversation correlation is spread across the sample; secondary stratum: question type. Allocation is proportional by largest remainder, with a minimum of one question per type present in a conversation. Seed 20261018 for all studies. Committed subsets store question-level records only; conversations are joined at load time from the local full dataset, and raw benchmark text is not redistributed.

F Statistical Procedures

Condition comparisons are paired by question. Exact McNemar: with b questions correct only under condition A and c correct only under condition B, the two-sided exact p-value is the binomial tail $2\min(P(X \leq \min(b,c)), P(X \geq \max(b,c)))$ with $X \sim \text{Binomial}(b{+}c, 0.5)$ under the null. Paired bootstrap CI: 10,000 resamples of question indices; percentile interval of the per-question accuracy difference. Cohen's κ (Study 2E): $\kappa = (p_o - p_e)/(1 - p_e)$ over the verdict categories correct/incorrect/uncertain, computed on the seeded subsample of Study 2B that both judges re-score. Because questions within one conversation are correlated, we report per-type breakdowns alongside pooled numbers and treat pooled p-values as optimistic (Dietterich, 1998).

G Environment

Answer models: qwen2.5:7b (digest 845dbda0ea48, 4.7 GB) and gemma3:12b (digest b92b218582c5, 8.2 GB); embedding backend bge-m3:latest (digest 790764642607, 1024-dim), all served locally via Ollama with temperature 0. Hardware: single laptop GPU (8 GB VRAM); no distributed inference. Per-study timings and code commits are recorded in each summary.json (models.code_commit); Study 2A/2B ran under commit a0fbf66 (post-C8 fix), the strict-mode reruns are the reported numbers. Every run is followed by audit_embedding_dims.py to verify the embedding backend served the full run (no silent hash degradation).

Responsible NLP Checklist

Datasets. LongMemEval-s (Wu et al., 2025) and LoCoMo (Maharana et al., 2024), both publicly released for research. We use published question-level subsets (LongMemEval-s, LongMemEval oracle subset, LoCoMo) and join conversations at load time from the local full data; raw benchmark text is not redistributed in the artifact (dataset versions and licenses are recorded in the artifact manifest). Both benchmarks contain English conversational dialogues; the subset used here contains no multi-modal records.

Personal data. Both benchmarks are synthetic or licensed fictional dialogues; no personal data of the authors or annotators is used. Speaker names are pseudonyms internal to the benchmarks.

Models and compute. Answer models qwen2.5:7b (Yang et al., 2024) and gemma3:12b, embedding model bge-m3

(Chen et al., 2024), all served locally via Ollama with recorded digests. Total compute is single-GPU (laptop-class, 8 GB VRAM); the full pipeline runs in a few hours per study.

Intended use and misuse. The protocol is intended for diagnosing memory-system failures in research settings. It is not a leaderboard metric: attribution characterizes the concrete implementation family under test (rule-based analyzer plus dense retrieval), and stage-boundary numbers should not be quoted as properties of other architectures. Automated attribution may mislabel cases where gold evidence is ambiguous; we therefore report E0_evidence_insufficient and eval_semantic_uncertain separately instead of forcing a failure label.

Anonymization. The public artifact will be anonymized for review (author-identifying metadata stripped; a separate anonymous branch is prepared by scripts/make_anonymous_branch.sh, with a verification grep and a fresh-repo push recipe for full history anonymization); model-server endpoints and paths are scrubbed.